

Brownian Motion, Bridges, and Bayesian Inference in Phylogenetic Tree Space

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November 2025

These notes are based on the paper [Woodman and Nye \[2025\]](#) plus some additional background that I have provided.

1 Background

The goal of phylogenetics is to use currently available information about a collection of species to try to infer the historical evolutionary tree, the sequence of shared common ancestors and speciation events that led to these modern day species. Typically we are interested in modeling both the *topology* of the evolutionary tree, the sequence of speciation events that separated species, as well as its *metric structure*, the amount of time passing between these speciation events. Historically, this was done with a variety of morphological features, from skeletal structures to plumage patterns, but since the discovery of DNA, this genetic information has played an increasing predominant role.

Given the DNA sequences for a specific gene shared across a collection of species, it is not too difficult to see how one might reconstruct the evolutionary tree: simply specify a model of mutation and select the evolutionary tree that is most likely to be obtained under this probabilistic model. For example, the simplest model, the Jukes-Cantor model, just assumes that each base pair $\{A, C, T, G\}$ in DNA mutates to another at rate μ uniformly at random. But as genomic sequencing improves and more genetic information becomes available, things become more difficult. For one, running the same analysis on two different gene sequences might produce significantly different tree topologies. Hence it becomes necessary to try to develop procedures that can somehow aggregate information across different genes.

The simplest approach is just concatenation, which treats all the available genetic information as one long string and running the above maximum-likelihood procedure. Beyond this, there are consensus methods which run the MLE procedure individually on each gene first and then uses the result collection of phylogenetic trees as inputs to a secondary procedure. For one, ASTRAL [Mirarab et al. \[2014\]](#) attempts to find the evolutionary tree which "overlaps" the most with the collection of gene trees by solving an optimization problem:

$$\hat{T} = \arg \max_T \sum_{g \in \mathcal{G}} d_{quartet}(g, T)$$

While many of these consensus methods have been proven to be asymptotically consistent, they often lack nice finite-sample guarantees. Even when we can develop these finite-sample bounds, statistical analysis is sort of ad-hoc because these models are not built on top of well-understood statistical theory.

We would like to somehow apply standard statistical theory, like concentration of the sample mean, PCA, and Bayesian inference models, but it is just difficult in a discrete space like this without much geometry. To get a flavor for what goes wrong, I'll start by discussing a traditional approach using Frechet means, and then move on to the new paper.

1.1 Frechet Means

Recall that in $\mathbb{R}^d$ the sample mean $\bar{x}$ of data points $\{x_i\}_{i=1}^n$ has the variational characterization:

$$\bar{x} = \arg \min_{c \in \mathbb{R}^d} \sum_{i=1}^n \|x_i - c\|_2^2$$

This is just a consequence of the bias-variance decomposition of squared error. Likewise, we can extend this to the population level. Given a probability distribution μ on $\mathbb{R}^d$ we know:

$$\mathbb{E}_{X \sim \mu} X = \arg \min_{c \in \mathbb{R}^d} \mathbb{E}_{X \sim \mu} \|X - c\|_2^2$$

The previous equation is really just a special case where μ is the empirical distribution of $\{x_i\}$. The *Frechet mean* is a natural extension of the ordinary Euclidean mean to a general metric space $(\mathcal{X}, d)$. Namely, in this new space we can define the sample mean as:

$$\bar{x} := \arg \min_{c \in \mathcal{X}} \sum_{i=1}^n d(x_i, c)^2$$

and likewise at the population level. If the space $(\mathcal{X}, d)$ is nice enough, the Frechet mean exists, is unique, and has the similar nice concentration properties to the sample mean in Euclidean space.

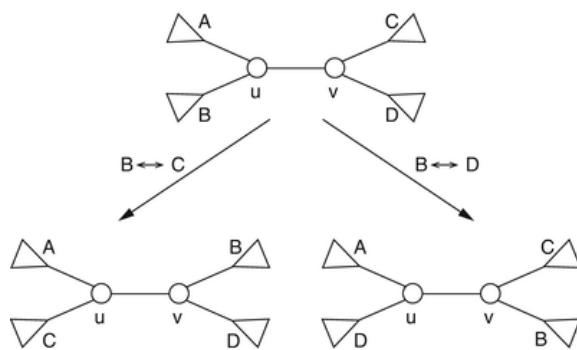


Figure 1: An example of two possible NNI moves.

Theorem 1 (Properties of Empirical Frechet Mean) *If $(\mathcal{X}, d)$ is a $CAT(0)$ space and μ has finite second moment, then the Frechet mean of μ exists and is unique [Sturm \[2003\]](#). Moreover if μ is sufficiently concentrated then the empirical Frechet mean of iid samples $X_i \sim \mu$ is concentrated around μ [Brunel and Serres \[2025\]](#).*

1.2 Introduction to BHV space

One common way of defining a metric on the space of trees is to count the number of "local" changes it takes to go from one tree to another. An example of such a change is the *nearest neighbor interchange* (NNI) which permutes the four subtrees adjacent to a given edge in an (unrooted) binary tree. Figure 1 gives an example. In the rooted case, this is equivalent to the standard tree rotations found in the likes of AVL trees.

In this picture, the space of phylogenetic trees is just an undirected graph: each vertex is a different tree and an edge connects two trees if they differ by an NNI move. One issue with this space is that shortest paths are not unique.

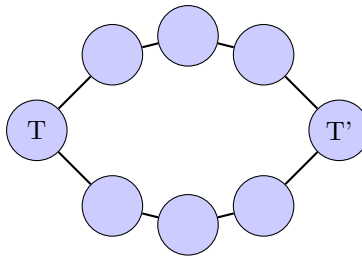


Figure 2: Nonunique paths in the discrete space of phylogenetic trees. Note both the upper center vertex and the lower center vertex would be a Frechet mean of $\{T, T'\}$

It is not hard to see that in such a discrete space, the Frechet mean is not unique, and in general two different means may be far apart. This is problematic because we would like to produce a single estimate $\hat{T}$ of the true evolutionary tree and study its asymptotic properties. The Billera-Holmes-Vogtmann (BHV) space [Billera et al. \[2001\]](#) was in a sense *designed* to solve this uniqueness issue.

Heuristically, the space BHV $\mathcal{T}_n$ of phylogenetic trees on n leaves is just a collection of orthants $\mathbb{R}_{\geq 0}^{N-3}$, one for each distinct tree topology. Each coordinate corresponds to the length of a particular internal edge, and two orthants are glued together on the lower-dimensional face corresponding to coordinate i if they differ by a single NNI move at the corresponding internal edge e_i . Hence each codimensional-1 face intersects exactly three n -cubes. Part of the space $\mathcal{T}_5$ is shown in Figure 3.

Theorem 2 *The space $\mathcal{T}_n$ is a $CAT(0)$ space. In particular, it has unique geodesics.*

As a corollary, we see the empirical Frechet mean always exists. Moreover, if we restrict our branch lengths to be all bounded by some constant C then $\mathcal{T}_n$ is compact and we get strong concentration around the population Frechet mean.

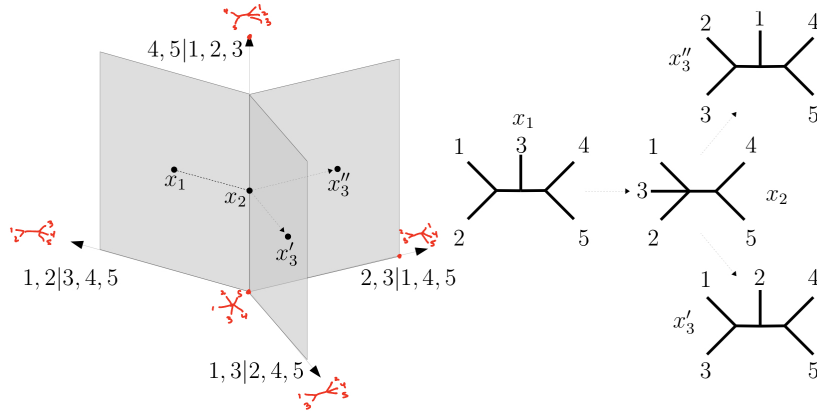


Figure 3: Part of the space BHV_5 . Here we show a few orthants glued together at their lower-dimensional face.

2 Brownian Motion in BHV space

The Fréchet mean $\bar{x}$ gives a model for the "center" of our collection of trees. There are analogous ways of defining its "dispersion". For one, we could look at the sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n d_{BHV}(x_i, \bar{x})^2$$

There's a whole literature on central limit theorems for Fréchet means, for one [Hotz et al. \[2022\]](#) or for stratified spaces in particular [Mattingly et al. \[2023\]](#).

A more parametric approach to summarizing a collection of trees would be to assume they are generated from some parametric model and to attempt to infer the parameters of this model. Of course, the classical choice for such a model would be the Gaussian. There are several nonequivalent models of what a Gaussian "should be" on such a space, and of course developing a notion of Brownian motion is one such way of generating one.

The paper [Nye \[2019\]](#) originally constructs BM on cubical complexes, of which BHV_N is a special case. Let $I = [0, 1]$. A cubical complex is a collection of hypercubes I^n glued together along their lower-dimensional faces: a simplicial complex except we use cubes instead of triangles. We say such a complex is n -dimensional if it contains an n -cube I^n and no higher dimensional cubes.

Heuristically, within the interior of a given n -cube, it is clear B_t should just evolve according to a standard n -dimensional Brownian motion. When B_t hits a codimensional-1 face f , we want B_t to choose a new n -cube uniformly at random from those adjacent to f and continue our standard Brownian motion. Namely, B_t will have two parts essentially:

- A reflected brownian motion in I^n which dictates our motion in the interior of the current n -cube.
- A discrete random walk on bipartite graph $G_{\mathcal{X}}$ with a vertex for each n -cube and each

$(n-1)$ -cube and an edge (u, v) if v is a face of u . This walk dictates the transitions between the different n -cubes of $\mathcal{X}$.

The main technical issue is that when we cross f , we will tend to recross it infinitely many times for every positive window of time afterwards before drifting away, so we have to be careful how we define our discrete random walk.

To formalize this, we start by defining a Brownian motion B_t on the hypercube I^n , which is relatively straightforward. Simply let each coordinate evolve independently as a standard Brownian motion on $[0, 1]$ with reflecting boundary conditions. For later use, we show that we can reasonably make sense of the sequence of faces B_t hits. Namely, given a continuous path $\eta : [0, 1] \rightarrow I^n$ let $\pi(\eta)$ be defined as the time-ordered sequence of codimensional-1 faces $(f_1, f_2, \dots)$ that η meets with consecutive repeats removed (so $f_i \neq f_{i+1}$). We must show there are not too many transitions between different faces:

Lemma 1 *If $x_0 \in I^n$ does not lie in a codimensional-2 face then $\pi(\eta)$ is well-defined and finite a.s. with respect to the distribution of B_t .*

This essentially shows us that the discrete walk corresponding to a continuous Brownian motion path is a.s. well-defined and just a standard simple random walk path on our undirected graph $G_{\mathcal{X}}$.

Proof 1 (Proof Outline) *The only way for B_t to hit a codimensional-2 face is if two coordinates (B_t^i, B_t^j) hit $(0, 0)$. But planar brownian motion hits the origin with probability zero, so $\pi(\eta)$ is well-defined.*

To prove $\pi(\eta)$ is a.s. finite, restrict to paths η that stay distance at least ρ from a codimensional-2 face. Show such paths a.s. have $\pi(\eta)$ finite, as they must travel at least distance ρ between faces, and Brownian motion cannot travel "that far" in finite time. Then send $\rho \rightarrow 0$.

Now we define our Brownian motion in terms of our reflected Brownian motion W_t on I^n as:

$$\mathbb{P}(B_t \in A) = \sum_{\omega} p(\omega) \cdot \mathbb{P}(W_t \in \mathcal{P}(A \cap \hat{C}^\omega)) \quad p(w) = \prod_i \frac{1}{\deg(v_i)} \quad (1)$$

where the sums essentially sum over all discrete random walk paths $(u_0, f_1, u_1, \dots, f_k, u_k)$ with non-repeating faces (e.g. $f_i \neq f_{i+1}$) and $\hat{C}^\omega$ is all continuous paths η whose corresponding discrete walk is ω . Lastly $\mathcal{P}$ is the projection map. If W_t has drift zero, covariance $\sigma^2 I_n$, and $W_0 = x_0$, we can view our Brownian motion B_t as parameterized by the same mean and dispersion parameters x_0, σ^2 .

2.1 Using Brownian Motion for Inference

We now have a reasonable model for Brownian motion, parameterized by a mean x_0 and a dispersion σ^2 that we would like to use to help us understand a tree-generating process. Namely, if we have sampled trees $\{x_i\}_{i=1}^n \in BHV_N$, we would like to try to use them to estimate x_0, σ^2 . The Bayesian approach would put some kind of prior π on these parameters and then use our samples $\{x_i\}$ to estimate the posteriors.

$$\mathbb{P}(x_0, \sigma^2 | \{x_i\}) \propto \pi(x_0, \sigma^2) \cdot \prod_{i=1}^n \mathbb{P}_{B_1}(x_i | x_0, \sigma^2)$$

where $\mathbb{P}_{B_1}$ is the distribution of our Brownian motion $B_t(x_0, \sigma^2)$ at time $t = 1$. Of course, the distribution of B_1 is intractable. Even though we have Equation (1), its not easy to understand the distribution of the reflected random walk.

Even calculating normalizing constants in this space for relative simple distributions can be challenging because you have to constantly keep track of how many orthants you are intersecting. For example, in BHV_4 the area of the ball $B_{BHV}(x_0, \epsilon)$ is $3\epsilon - d_{BHV}(x_0, 0) \wedge \epsilon$, seen in Figure 4.

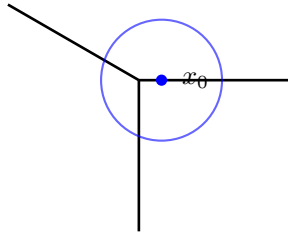


Figure 4: An ϵ -ball in BHV_4

This problem is only exacerbated near lower-dimensional faces, where balls intersect many orthants in complicated ways. Thus we approximate B_1 in two steps:

1. We approximate our continuous walk by a discrete walk.
2. If the steps of our discrete walk are small enough, they typically cross at most one face (of codimension-1). In this case, the distribution of the steps is much easier to understand.

2.1.1 Discrete Approximation using Gaussian Via Geodesic Firing

Suppose that x_0 is in the interior of an n -cube. Given dispersion parameter τ^2 define the *gaussian via geodesic firing* $GGF(x_0, \tau^2)$ via:

- Sample a direction $v \sim N(0, \tau^2 I_{N-3})$.
- Follow the unique geodesic starting at x_0 in direction v .
- If you intersect a codimensional-1 face, choose one of the three adjacent n -cubes uniformly at random and continue within that cube in the same direction.
- Continue until you travel total distance $|v|$.

If we iterate this process, we obtain an analog of the simple random walk with Gaussian increments. Unsurprisingly, we get the following central limit theorem.

Lemma 2 *If $y_0 \equiv x_0$ and we sample $y_i | y_{i-1} \sim GGF(y_{i-1}, \sigma^2/m)$ let $W(x_0, \sigma^2; m)$ denote the distribution of y_m . Then $W(x_0, \sigma^2; m) \Rightarrow B_1(x_0, \sigma^2)$ as $m \rightarrow \infty$.*

Hence our first approximation:

$$\mathbb{P}(x_0, \sigma^2 | \{x_i\}) \propto \pi(x_0, \sigma^2) \cdot \prod_{i=1}^n \mathbb{P}_{B_1}(x_i | x_0, \sigma^2) \approx \pi(x_0, \sigma^2) \cdot \prod_{i=1}^n \mathbb{P}_W(x_i | x_0, \sigma^2; m)$$

Of course, $\mathbb{P}_{W(x_0, \sigma^2; m)}(x_i)$ is still intractable due to the combinatorial nature of the discrete walks and the complicated reflection process. The key idea is that if we choose the number of steps m to be large enough, each step in our discrete walk crosses at most one face. In this case, the distribution of the individual steps of W are more tractable. Hence we ought to condition on the individual points y_i in our discrete walk, which is the content of our next section.

2.1.2 Marginalization over bridges

As discussed previously, we can essentially observe:

$$\mathbb{P}_W(x_i | x_0, \sigma^2; m) = \int \mathbb{P}_W(\mathbf{y}, x_i | x_0, \sigma^2; m) \mathbb{P}_W(d\mathbf{y}) = \int \prod_{j=1}^m f_{GGF}(y_j | y_{j-1}; \sigma^2/m) \mathbb{P}_W(d\mathbf{y})$$

where the integral is over the distribution of *bridges* from x_0 to x_i . Like we mentioned above the distribution of f_{GGF} can easily be approximated when m is large enough. We then just need to sample many bridges and we can approximate $\mathbb{P}(d\mathbf{y})$ by its empirical distribution. Overall then if we sample M bridges $\mathbf{y}^1, \dots, \mathbf{y}^M$, we get:

$$f(x_0, \sigma^2 | \mathbf{x}) \propto \frac{1}{M} \sum_{\ell=1}^M \pi(x_0, \sigma^2) \cdot \prod_{i=1}^n \prod_{j=1}^m f_{GGF}(y_{i,j}^\ell | y_{i,j-1}^\ell, \sigma^2/m) \quad (2)$$

Again this only calculates the posterior up to the proportionality constant, but we can always use MCMC to sample from this posterior. All that is left to do is sample a bridge $\mathbf{y}_i$ from x_0 to x_i of this discrete random walk $W(x_0, \sigma^2; m)$.

2.1.3 Sampling Bridges

To sample bridges, we take inspiration from the standard Brownian bridge. There recall that if we have a Gaussian random walk $Y_0, Y_1, \dots, Y_m$ with Y_0 then:

$$Y_j | \{Y_{j-1} = y_{j-1}, Y_m = x^*\} \sim N\left(y_{j-1} + \frac{1}{m-j+1}(x^* - y_{j-1}), \frac{m-j}{m-j+1} \cdot \frac{\sigma^2}{m} I_{N-3}\right)$$

Namely, we follow the geodesic from y_{j-1} to x^* for time $\frac{1}{m-j+1}$ and add a small gaussian perturbation. Likewise, for our discrete random walk we can follow a similar procedure to simulate a bridge. Given $y_0 = x_0$ and conditioning on $W(x_0, \sigma^2; m) = x^*$ we run the following:

1. Construct the geodesic Γ_{y_{j-1}, x^*} and follow it distance $\frac{1}{m-j+1}$ to get μ_j .
2. Sample $y_j \sim GGF(\mu_j, \tau_{j,m})$ where $\tau_{j,m} = \frac{m-j}{m-j+1} \cdot \frac{\sigma^2}{m}$.
3. If Γ_{y_{j-1}, y_j} cross a face of codimension greater than one, restart.

The last step is because the GGF density is intractable to calculate at such lower dimensional faces. Again this is only gives approximately the correct distribution.

In the rest of the paper, the author proposes a few heuristics that can improve the algorithm. For one, step (3) can cause the algorithm to fail quite frequently, so the paper proposes several adjustments that can be made to reduce the failure rate. Also, as we discussed above the approximation of f_{GGF} becomes worse the closer we get to lower-dimensional faces, so the author also proposes some methods of perturbing the walk in a way that stays far enough away from such faces.

3 Final Remarks

These parameters give us a new notion of the "mean" and "dispersion" of our sample of trees. Moreover, once we learn the parameters, we can use them to generate new potential trees for downstream usage.

Some natural questions I had after reading the article:

- Will the MLE for the center x_0 of our brownian motion not just be the Fréchet mean? At least, we could assume it was and then try to learn the variance parameter σ^2 possibly?
- The author does not really discuss it, but I imagine we can develop analogs of the Brownian motion where we choose the reflecting brownian motion W_t on I^n to be non-standard. Namely, assume W_t has drift μ and covariance Σ more generally rather than the case $\mu = 0, \Sigma = \sigma^2 I$ discussed in this paper. The only technical difficulty is that we would have to find a reasonable way of adjusting the discrete random walk to no longer uniformly choose an n -cube adjacent to the current face. e.g. surely if $\mu = (1, 0)$ and we exit out of the right side of I^2 then it ought to be more likely to leave the current cube than return due to the drift.

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